

CHAPTER 11

Crack Spread Trading

Trading the refining margin — the spread between crude and the products it becomes.



Trading the margin, not the barrel

The **3-2-1 crack spread** — three barrels of crude yielding two of gasoline and one of distillate — is the standard proxy for a refiner's margin. A trader can buy or sell that margin directly, without ever owning a refinery or a barrel.

- **A spread, not a flat price.** The difference between product prices and crude isolates the refining margin from the direction of oil itself.
- **Why it exists.** It lets refiners lock in profit and traders bet on whether refining will be lucrative — independent of where crude goes.
- **The bridge between chapters.** Crack-spread trading joins the crude market to the product markets, the two priced against each other.

THE 3-2-1 YIELD



What a crack spread is



The refiner's margin

The gap between what a refiner pays for crude and what it earns selling the refined products — its theoretical gross margin.

The name

To "crack" crude is to break it into lighter products. The spread measures the value created by that transformation.

Quoted per barrel

Expressed in \$/bbl. A positive crack means refining is profitable; a negative crack means products are worth less than the crude.

A market of its own

Because it can be bought and sold as a single position, the crack spread is a tradable instrument, not just an accounting figure.

BENCHMARK RATIOS

The 3-2-1 and its cousins

3-2-1



The standard benchmark — three crude produce two gasoline and one distillate, roughly a typical refinery's yield.

5-3-2



A finer approximation that some refiners use to mirror their product slate more closely.

1-1-1



Single cracks isolate one product — a gasoline crack or a heating-oil crack against one barrel of crude.

× 42

Convert to one unit. Products are quoted in cents per gallon, so each is multiplied by 42 gallons to reach \$/bbl before the spread is calculated.

WORKED EXAMPLE

How the 3-2-1 is calculated



- 1 Convert to \$/bbl.** $\times 42$ gal. Gasoline $\$2.50/\text{gal} \rightarrow \$105/\text{bbl}$; distillate $\$3.00/\text{gal} \rightarrow \$126/\text{bbl}$.
- 2 Value the products.** $2 \times \$105 + \$126 = \$336$ combined product value.
- 3 Subtract the crude.** $3 \times \$90 = \270 , so gross margin = $\$336 - \$270 = \$66$.

$\$66 \div 3$ barrels
just above the typical \$10–20 band

\$22/bbl

WHAT THE NUMBER MEANS

Reading the crack



● **~\$15**
normal 3-2-1 today

● **~\$83–88**
record diesel crack, 2022

A positive crack signals profitable refining; the wider it runs, the harder refiners push to capture it before it closes.

Buying and selling the crack

Sell the crack

The refiner's natural hedge — locks in the margin

SELL ■ Product futures (gasoline + distillate)

BUY ■ Crude futures

Buy the crack

The trader's bullish-margin position

BUY ■ Product futures (gasoline + distillate)

SELL ■ Crude futures

One position, three legs

A 3-2-1 crack is executed across multiple contracts at once, but managed as a single spread with a single margin figure.

Exchange-listed cracks

Exchanges list ready-made crack-spread contracts, so the whole position can be traded in one click rather than leg by leg.

Why refiners hedge the crack

Their real exposure

A refiner cares less about the flat price of crude than about the gap between crude and products — that gap is its actual profit.

Locking in profit

By selling the crack, a refiner fixes its margin months ahead, protecting against products falling or crude jumping.

Timing input and output

Refiners buy crude in one month and sell products later, so the crack also hedges the time gap between input and output.

Run decisions

When the crack is wide, refiners run hard and hedge it in; when it turns negative, they cut runs rather than refine at a loss.

What moves the crack spread



The price lag

Products follow crude with a delay. After a crude spike, refiners selling at new product prices briefly capture a wider crack.



Demand & season

Strong gasoline demand in summer or diesel demand in winter lifts the relevant crack independently of crude.



Refining capacity

Outages, run cuts and seasonal maintenance tighten product supply and widen cracks; new capacity narrows them.



Feedstock advantage

A refiner buying cheaper crude relative to its products — WTI versus Brent-priced fuels — earns a structurally wider crack.

WHEN THE CRACK BLOWS OUT

\$83

record diesel crack, 2022

When Europe lost Russian diesel in 2022, the diesel crack spiked to roughly **\$83 a barrel** — and touched a documented **~\$88** at the autumn peak. A healthy 3-2-1 sits near \$10–20; disruptions push it past \$30–40, and a true squeeze far beyond.



When correlation breaks — the 2026 squeeze

After the Iran conflict closed in on Hormuz in early 2026, gasoline and distillate prices outran crude and middle-distillate cracks hit all-time highs — a live replay of every driver in this chapter.

+180%

3-2-1 crack surge

WTI 3:2:1 up ~\$30/bbl after the conflict began

20 → 3.8

Hormuz flows, mb/d

Crude, NGL & product loadings, Feb → April 2026

−10.1

Supply shock, mb/d

Global supply fell to ~97 mb/d in March 2026

The WTI edge

US Gulf Coast refiners bought cheaper WTI while selling products priced off Brent, turning the widening WTI–Brent gap straight into margin.

The lesson

The crack can move violently and independently of flat price — exactly why it is hedged and traded on its own.

Sources: IEA Oil Market Report (Apr–May 2026); Dallas Fed Energy Indicators (Mar 2026).

Trading the crack as a strategy

A view on refining, not oil

A crack trade profits from a tightening or loosening product market without a flat-price bet on crude.

Lower directional risk

Because both legs move with oil, much of the flat-price risk cancels out, leaving a cleaner bet on the margin itself.

Seasonal & event plays

Traders position for summer gasoline, winter diesel, or a refinery outage — all expressed through the crack.

Liquidity and the legs

Cracks are most reliable where both the crude and product contracts are deeply liquid, keeping execution tight.

The risks of crack trading

Not a perfect hedge

The 3-2-1 only approximates a real refinery's slate. The actual margin can drift from the benchmark — this gap is basis risk.

Correlation can break

In a shock, products and crude can decouple sharply. That is the opportunity in crack trading — but also its danger.

Legging risk

Executing the crude and product legs separately leaves a brief window where prices can move against you.

Still leveraged

A crack is a spread, but each leg is a leveraged futures position. Disciplined margin and risk management remain essential.

CONCLUSION — THE MARGIN MADE TRADABLE

Learn to read and trade the crack, and you isolate refining value itself.

The refiner's world

One number decides whether refining is profitable — the spread between crude and products.

A hedge and a bet

Refiners sell it to lock in margin; traders buy and sell it to express a view on refining.

Crude meets products

Crack trading joins the two markets through the futures and spread mechanics of the course.

Sources

- **U.S. EIA — “3:2:1 Crack Spread” & “An introduction to crack spreads”**
Standard 3-2-1 formula, WTI / RBOB / NY Harbor ULSD definition.
- **CME Group — Trading Crack Spreads; NY Harbor ULSD crack**
Exchange-listed crack contracts and hedging mechanics.
- **EIA This Week in Petroleum (2022)**
2022 distillate crack record (~\$83–88/bbl) during the diesel squeeze.
- **IEA — Oil Market Report, April & May 2026**
2026 Hormuz supply shock; record middle-distillate cracks.
- **RBN Energy — 3-2-1 Crack Spread; Brent–WTI Spread**
Live refining-margin and transatlantic-spread benchmarks.
- **Dallas Fed — Energy Indicators, March 2026**
WTI–Brent differential and Gulf Coast refiner margin context.

Worked figures are illustrative (\$2.50/gal gasoline, \$3.00/gal distillate, \$90/bbl crude → \$22/bbl crack). Market levels are approximate and verified against the sources above as of June 2026.